PROJECT (Group work)

Instructions: In your groups, implement an information system of your choice following the guidelines given below:

To help you put into practice the concepts, methodologies and techniques learnt in the Software Engineering unit, you will be required to design and develop a fully functional and complete information system.

The project, which should be done in your groups, should have the **front-end**, **Admin panel** and **database**.

The front-end is what ordinary users/customers/clients interact with on a day-to-day basis to perform their tasks on the system.

The admin panel is a user interface accessed by the administrator(s) of the system to perform administrative tasks like registering users, resetting passwords, adding stock, etc, etc.

The database is the store of data that interacts with both the front-end and admin panel and is created using your favorite Database Management System.

DELIVERABLE 1: PROJECT TITLE => PLANNING / REQUIREMENTS ANALYSIS PHASE (DUE

The first stage where you are expected to hand in the

i. Project title,
ii. a brief description of the system you intend to develop,
iii. group members and the tasks assigned to each member defined
iv. 5 Functional requirements and
v. 3 Non-functional requirements.

DELIVERABLE 2: DIAGRAMS AND MOCK-UPS => SYSTEMS ANALYSIS & SYSTEMS DESIGN PHASES (DUE DATE 2026)

The Front-end UI design & admin panel UI design are Graphical User Interface sketches done on paper or mock-ups/wireframes done by software tools like balsamiq, wireframesketcher,mockupui etc (https://balsamiq.com/wireframes, https://wireframesketcher.com, https://mockupui.com)

The Database schema is NOT the developed database. It is what you will refer to/use when creating the DB. It can be written or drawn and should capture the database name, table names, all attributes and their data types, the primary and foreign keys, plus any other relevant constraints on all your database tables.

You will be expected to submit the Flow chart diagram, Use case diagram, class diagram, E-R diagram, database schema and User interface mockups/sketches.

DELIVERABLE 3: MID-SEMESTER MARKING => DEVELOPMENT & DEBUGGING PHASES (DUE DATE 2026)

Progress marking of:

Developed/created a database in a DBMS of your choice.

Well-designed and developed Front-end.

Well-designed and developed Admin panel.

Connection of the front-end to the Database.

Connection of the admin panel to the Database.

DELIVERABLE 4: FINAL PRESENTATION & MARKING => TESTING, DEPLOYMENT & MAINTENANCE PHASES (DUE DATE)

Presentation and marking of the complete working system that has all the functionalities implemented.

Marks will be given for:

DB: identification of all entities and attributes, fully normalized database, correct relationships & cardinalities.

Uniformity and ease of use/User friendliness.

Data processing: ADD, DELETE, UPDATE/EDIT, BROWSE/SEARCH.

Help facilities/files and relevant error messages.

Transactions/Business Logic: The Main transaction MUST be working (i.e reservation, booking, scheduling, etc).

Levels of Access/User profiles->Admins/users/super admins/supervisors, Login, logout, user registration, forgot password.

Reports (Crystal Reports or ireports/Jasper reports, etc): layout/Structure of the reports, Date, company name & logo on the reports, Report filtering/sorting.

Use of frameworks, libraries, APIs and plugins (Bootstrap, Font Awesome, M-Pesa API, etc, etc).

Form validation, error-handling/Exception handling, key trapping and recovery, Module Integration.

Creativity: unique features, effort, and ingenuity.

Well-designed ('beautiful') GUI, Intuitive design.

Coding best practices: Commenting, naming conventions, code simplicity, Portability, Modularity.

Good presentation skills, Audibility, understanding of the problem, and logical solution.

The solution/system presented meets objectives and solves real-world problem(s).

System processes data.

Packaging and deployment: Installation wizards/exes, live system hosted on web server.

User manual, documentation.

Audit logs/audit trails to help keep track of what happened on your system and was done by whom.

Security (Application & Database security): Password hashing, encryption, Use of prepared statements to prevent SQL injection, two-factor authentication (2FA), etc

Punctuality and good use of allocated time.